# Reviewer Pack — Noncrossing Partition Tree Heights and AC^0 Circuit Size

Entry 411d9e5b8351 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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## Noncrossing Partition Tree Heights and AC^0 Circuit Size

**Entry ID**: `411d9e5b8351` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-25 20:09:23 UTC

### 1. Conjecture

**Field A** (mathematical branch): Enumerative Combinatorics (Noncrossing Partitions) **Field B** (complexity object): Complexity Theory: AC^0 Circuit Complexity

**Statement**:

> ['For all instances of Boolean functions f with n variables, the height h(n) of a minimal noncrossing partition tree representing f is upper bounded by the size of the smallest AC^0 circuit computing f.', 'Specifically, there exists a constant C such that for all n ≤ 40, h(n) ≤ 2^(C * log^2 n).', 'For every function with polynomially-bounded noncrossing partition tree height, its computation can be achieved by an AC^0 circuit of similar size.']

**Rationale (proposer's reasoning)**:

> ['Noncrossing partitions have been used in various combinatorial problems, but their relationship to circuit complexity has not been extensively explored.', 'This conjecture suggests that noncrossing partition trees could provide a natural way to represent Boolean functions with efficient circuits.', 'If proven true, it would offer a new approach to understand the complexity of AC^0 circuits and potentially lead to improvements in circuit lower bounds.']

**Taxonomy category**: `ENUMERATIVE_COMBINATORICS_NONCROSSING_PARTITIONS` (status at proposal time: )

### 2. Pre-registration (Popper-style)

**Hash (SHA-256 prefix)**: `ac9a8daf11064f07`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

**Acceptance criterion**:

> For n ≤ 40, if the height of a minimal noncrossing partition tree h(n) for any Boolean function f is less than or equal to 2^(C * log^2 n), and this condition holds true for at least 24 out of 30 random seeds.

## 3. Barrier filter (F1)

**Final**: PASS

| Barrier | Verdict | Confidence | LLM₁ | LLM₂ |
|---|---|---|---|---|
| RELATIVIZATION | SAFE | 1.00 | UNCERTAIN | SAFE |
| NATURAL_PROOFS | SAFE | 0.50 | UNCERTAIN | UNCERTAIN |
| ALGEBRIZATION | SAFE | 0.50 | UNCERTAIN | UNCERTAIN |
| KARP_LIPTON | SAFE | 0.50 | UNCERTAIN | UNCERTAIN |

## 4. Novelty audit

**Verdict**: NOVEL against 2 hits across arXiv + Semantic Scholar + ECCC

**Search queries** (3): - "noncrossing partition tree" AND "AC^0 circuit complexity" - "minimal noncrossing partition tree" AND "circuit size" - "Boolean function" AND height AND "AC^0 circuit"

**Top relevant hits considered**: - [s2:10.1007/s00236-025-00489-9] Row-column combination of Dyck words - [s2:b24f4cd7d53c0566b3f07a216b160e1a06c25ccc] Addressing signal electromigration ( EM ) in today ' s complex digital designs

## 5. Empirical test harness

**Multi-seed protocol**: 5 seeds (11, 23, 37, 53, 71)
**Sandbox**: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

### 5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def noncrossing_partition_tree_height(f):
        n = len(f)
        if n == 1:
            return 1
        height = 0
        while f:
            new_f = []
            for i in range(len(f) // 2):
                a, b = f[2*i], f[2*i+1]
                if a != b:
                    new_f.append(1)
                else:
                    new_f.append(a)
            f = new_f
            height += 1
        return height
```

```
    def ac0_circuit_size(f):
        n = len(f)
        if n == 1:
            return 1
        size = 0
        while f:
            new_f = []
            for i in range(len(f) // 2):
                a, b = f[2*i], f[2*i+1]
                if a != b:
                    new_f.append(1)
                else:
                    new_f.append(a)
            f = new_f
            size += 1
        return size

    n_values = [5, 10, 15, 20, 30, 40]
    results = []

    for n in n_values:
        for _ in range(5):
            f = generate_boolean_function(n)
            height = noncrossing_partition_tree_height(f)
            size = ac0_circuit_size(f)
            if height > 2**(C * math.log2(n)**2):
                return {
                    "metric_name": "height vs circuit size",
                    "metric_value": height,
                    "instances_tested": 1,
                    "conjecture_holds": False,
                    "counterexample": f"n={n}, h(n)={height}, AC^0 size={size}"
                }
            results.append((height, size))

    mean_height = sum(h for h, _ in results) / len(results)
    mean_size = sum(s for _, s in results) / len(results)

    return {
        "metric_name": "height vs circuit size",
        "metric_value": mean_height,
        "instances_tested": len(results),
        "conjecture_holds": all(h <= 2**(C * math.log2(n)**2) for n, h in results),
        "counterexample": ""
    }

if __name__ == "__main__":
    if not sys.argv[1:]:
        seeds = [2**i + 1 for i in range(5, 30)]
    else:
        seeds = list(map(int, sys.argv[1:]))

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
```

```
        results.append(trial_result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std=0.0 support_fraction={support_fract
    else:
        for r in results:
            if not r["conjecture_holds"]:
                counterexample = r["counterexample"]
                first_failing_seed = seed
                break

        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_fa
```

## 6. Per-seed results

*(no seed-level data — test crashed before producing TRIAL: lines)*

## 7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e7580757.py", line 95, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e7580757.py", line 66, in run_trial
    if height > 2**(C * math.log2(n)**2):
                    ^
NameError: name 'C' is not defined
```

## 8. Critic adversarial review

**Critic verdict**: CHALLENGE

**Critic reasoning**:

> skipped: test crashed before producing data

## 9. Final verdict & safety rail

**Verdict**: INCONCLUSIVE

**Reasoning**:

> The test code crashed due to an undefined constant ‘C’, which prevented it from producing any data. As the critic challenged the conjecture and the pre-registered support condition was not unambiguously met, the verdict is INCONCLUSIVE. | next: Define the constant ‘C’ in the test code and rerun the test to verify if the conjecture holds for at least 24 out of 30 random seeds.

## 11. Audit log (LLM calls)

**Total LLM calls**: 9

| # | Phase | Provider | Model | Tokens in | out | Latency (ms) |
|---|---|---|---|---|---|---|
| 1 | propose | ollama_remote | glm4:latest | 0 | 0 | 11575 |
| 2 | preregistration | ollama_remote | glm4:latest | 0 | 0 | 5594 |
| 3 | novelty | ollama_remote | glm4:latest | 0 | 0 | 4731 |
| 4 | novelty | ollama_remote | glm4:latest | 0 | 0 | 5866 |
| 5 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 15113 |
| 6 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 8555 |
| 7 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 11427 |
| 8 | test_gen | ollama_remote | qwen2.5-coder:7b | 0 | 0 | 9694 |
| 9 | judge | ollama_remote | glm4:latest | 0 | 0 | 9916 |

**Totals**: 0 input tokens, 0 output tokens, ~$0.0000 cost, 82471 ms total latency. Provider mix: {'ollama_remote': 9}

*(full prompt+response transcripts available in `research/audit/411d9e5b8351.jsonl`)*

## 12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/411d9e5b8351.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

**Sandbox file**: `research/pvsnp_sandbox/test_*.py` (per-cycle)
**Replay tarball**: `research/replay/411d9e5b8351.tar.gz` (if generated)